

Get ready for bubbles

Now the submarine can move, start creating the bubbles for the player to pop. Each bubble will be a different size and move at a different speed.

- 4** Every bubble needs an ID number (so the program can identify each specific bubble), a size, and a speed.

```
from random import randint
bub_id = list()
bub_r = list()
bub_speed = list()
MIN_BUB_R = 10
MAX_BUB_R = 30
MAX_BUB_SPD = 10
GAP = 100
def create_bubble():
    x = WIDTH + GAP
    y = randint(0, HEIGHT)
    r = randint(MIN_BUB_R, MAX_BUB_R)
    id1 = c.create_oval(x - r, y - r, x + r, y + r, outline='white')
    bub_id.append(id1)
    bub_r.append(r)
    bub_speed.append(randint(1, MAX_BUB_SPD))
```

This creates three empty lists used to store the ID, radius (size), and speed of each bubble

Sets the minimum radius of the bubble to 10, and the maximum to 30

Sets the position of the bubble on the canvas

Picks a random size for the bubble, between the minimum and maximum values possible

This line of code creates the bubble shape

Adds the ID, radius, and speed of the bubble to the three lists



EXPERT TIPS

Bubble lists

Three lists are used to store information about each bubble. The lists start off empty, and information about each bubble is then added as you create it. Each list stores a different bit of information.

bub_id: stores the ID number of the bubble so the program can move it later.

bub_r: stores the radius (size) of the bubble.

bub_speed: stores how fast the bubble travels across the screen.



Don't forget to
save your work

